Homework reference: compact print edition

Joseph Bailey October 30, 2026

Problem statements and solutions for all exercises in HW1–HW4 are included. Statements are restated from Dembo and the HW4 handout, with notation normalized and all subparts retained. Corrections identified in the review are incorporated in this print copy; submitted originals are unchanged. Major repairs: the HW1 union bound and continuity notation; HW4 finite partial sums, common normalization, infinite-product limits, nonnegative zero-expectation criterion, and L^p cases. Other edits clarify notation and hypotheses.

Lookup index. These page references point into *this compact edition*. Exercise numbers remain the same; old homework page numbers should not be used here.

Need a model for	Exercise / method	Page
Probability bounds and continuity	HW1 1.1.4: disjointization, monotone sets	2
Prove a collection is a sigma-algebra	HW1 1.1.13: closure properties	2
Borel/product generators	HW1 1.1.21: prove both inclusions	3
Countable dependence on generators	HW1 1.1.22: a sigma-algebra argument	3
A uniqueness counterexample	HW1 1.1.33: generators need not be a pi-system	4
Sigma-algebras generated by variables	HW2 1.2.14: threshold events	4
Measurability from a generating class	HW2 1.2.20: preimages of half-lines	5
Small exceptional sets	HW2 1.2.30: decreasing tail unions	5
Pointwise construction of measurable functions	HW2 1.2.31: simple functions and limits	5
Support of a distribution	HW2 1.2.50: neighborhoods and CDF jumps	5
Extend an integral identity	HW3 1.3.7: standard machine	6
Moment bounds and event inequalities	HW3 1.3.21: Cauchy–Schwarz, indicators	6
Random indices and convergence	HW3 1.3.36–37: tail unions, typewriter example	7, 8
Expectations of infinite sums	HW4 1.3.65: partial sums plus DCT	8
Pairwise versus mutual independence	HW4 1.4.15: finite bijections, constraints	9
Independence and infinite products	HW4 1.4.18: finite products, continuity	9
L^p norms and simple approximation	HW4 parts 1–6: a.s. classes, Minkowski, DCT	10–11

Definitions to find instantly.

$$f = g \quad \mu\text{-a.e.} \iff \mu(\{f \neq g\}) = 0, \qquad f \in L^1 \iff \int |f|\, d\mu < \infty.$$

On a probability space, a.e. and a.s. mean the same thing. By contrast, $f_n \to f$ in L^1 means $\int |f_n - f|\, d\mu \to 0$. In Dembo: equality a.s., Definition 1.2.8, PDF p.19; the terminology link, p.20; integrability, Definition 1.3.2, p.32; L^q convergence, Definition 1.3.26, p.39.

Homework 1

Problem 1.1.4. Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space and $A, B, A_i \in \mathcal{F}$. Prove (a) if $A \subseteq B$, then $\mathbf{P}(A) \leq \mathbf{P}(B)$; (b) if $A \subseteq \bigcup_i A_i$, then $\mathbf{P}(A) \leq \sum_i \mathbf{P}(A_i)$; (c) if $A_i \uparrow A$, then $\mathbf{P}(A_i) \uparrow \mathbf{P}(A)$; (d) if $A_i \downarrow A$, then $\mathbf{P}(A_i) \downarrow \mathbf{P}(A)$.

Solution.

(a) Suppose $A \subseteq B$. $B = A \cup (B \setminus A)$. From countable additivity,

$$\mathbf{P}(B) = \mathbf{P}(A) + \mathbf{P}(B \setminus A).$$

From nonnegativity, $\mathbf{P}(A) \leq \mathbf{P}(B)$.

(b) Suppose $A \subseteq \cup_i A_i$. Similarly to (a), define disjoint sets $B_1 = A \cap A_1$, $B_n = A \cap A_n \setminus \left(\bigcup_{i=1}^{n-1} A_i\right)$. Then $B_i \subseteq A_i$ and $\cup_i B_i = A \subseteq \cup_i A_i$. Using countable additivity and nonnegativity,

$$\mathbf{P}(A) = \mathbf{P}(\cup_i B_i) = \sum_i \mathbf{P}(B_i) \leq \sum_i \mathbf{P}(A_i).$$

(c) Suppose $A_i \uparrow A$. Define disjoint sets $B_1 = A_1$, $B_n = A_n \setminus A_{n-1}$. Then $\cup_i^n B_i = A_n$ and $\cup_i B_i = \cup_i A_i = A$. Using countable additivity,

$$\lim_{n\to\infty} \mathbf{P}(A_n) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbf{P}(B_i) = \sum_{i=1}^{\infty} \mathbf{P}(B_i) = \mathbf{P}(A).$$

(d) Suppose $A_i \downarrow A$. Let $B_n = A_n^c$ and $B = A^c$. These are in $\mathcal{F}$ using closure under complements. $A_n \supseteq A_{n+1} \implies B_n \subseteq B_{n+1}$ and using De Morgan's laws:

$$\bigcup_{i=1}^{\infty} B_n = \left(\bigcap_{i=1}^{\infty} A_n\right)^c = A^c = B,$$

so $B_n \uparrow B$. Using (c), $\lim_{n\to\infty} \mathbf{P}(B_n) = \mathbf{P}(B)$ and because

$$\mathbf{P}(B_n) = \mathbf{P}(\Omega \setminus A_n) = 1 - \mathbf{P}(A_n) \text{ and } \mathbf{P}(B) = 1 - \mathbf{P}(A),$$

$\lim_{n\to\infty} \mathbf{P}(A_n) = \mathbf{P}(A)$ or $\mathbf{P}(A_n) \downarrow \mathbf{P}(A)$.

Problem 1.1.13. (a) Show that the intersection of any nonempty family of sigma-algebras on Ω, possibly uncountable, is a sigma-algebra. (b) For sigma-algebras $\mathcal{H} \subseteq \mathcal{G}$ and $H \in \mathcal{H}$, show that $\mathcal{H}^H = \{A \in \mathcal{G} : A \cap H \in \mathcal{H}\}$ is a sigma-algebra. (c) Show that $H \mapsto \mathcal{H}^H$ reverses inclusion, $\mathcal{H}^\Omega = \mathcal{H}$, $\mathcal{H}^\varnothing = \mathcal{G}$, and $\mathcal{H}^{H \cup H'} = \mathcal{H}^H \cap \mathcal{H}^{H'}$ for $H, H' \in \mathcal{H}$.

Solution.

(a) Let $\mathcal{F} = \cap_\alpha \mathcal{F}_\alpha$ be a intersection of uncountably many of σ-algebras.

i. Each $\mathcal{F}_\alpha \ni \emptyset$, so so does the intersection.

ii. If $A \in \mathcal{F}$, then $A \in \mathcal{F}_\alpha \ \forall \alpha$. Using closure under complements, $A^c \in \mathcal{F}_\alpha \implies A^c \in \mathcal{F}$.

iii. If $A_i \in \mathcal{F} \ \forall i$, then $A_i \in \mathcal{F}_\alpha$. Using closure under countable unions, $\cup_i A_i \in \mathcal{F}_\alpha \ \forall \alpha \implies \cup_i A_i \in \mathcal{F}$.

(b) i. All $\mathcal{H} \ni \emptyset$ since they are σ-algebras and $\emptyset \cap H = \emptyset$, so $\emptyset \in \mathcal{H}^H$.

ii. $\mathcal{H}$ is closed under compliments, any $A \in \mathcal{H}^H$, $A^c \cap H = H \setminus (A \cap H) \in \mathcal{H}$. So $A^c \in \mathcal{H}^H$.

iii. Take $A_i \in \mathcal{H}^H$. Then $(\cup_i A_i) \cap H = \cup_i (A_n \cap H) \in \mathcal{H}$ using the closure under countable unions of $\mathcal{H}$. So $\cup_i A_i \in \mathcal{H}^H$.

(c) Every $A \cap \Omega = A$, so $A \cap \Omega \in \mathcal{H} \iff A \in \mathcal{H}$, so $\mathcal{H}^\Omega = \mathcal{H}$. $\mathcal{H} \ni \emptyset$, so for every $A \in \mathcal{G}$, $A \cap \emptyset = \emptyset \in \mathcal{H}$. Thus $\mathcal{H}^\emptyset = \mathcal{G}$.

To see non-increasing with set inclusions, suppose $H_1 \subseteq H_2$. If any $A \in \mathcal{H}^{H_2}$ and $A \cap H_2 \in \mathcal{H}$, because $H_1 \in \mathcal{H}$ and $\mathcal{H}$ is closed under countable intersections,

$$(A \cap H_2) \cap H_1 = A \cap H_1 \in \mathcal{H},$$

so $A \in \mathcal{H}^{H_1}$ and $\mathcal{H}^{H_2} \subseteq \mathcal{H}^{H_1}$.

Because the mapping is non-increasing and each H, H' is a subset of their union,

$$\mathcal{H}^{H \cup H'} \subseteq \mathcal{H}^H \text{ and } \mathcal{H}^{H \cup H'} \subseteq \mathcal{H}^{H'} \implies \mathcal{H}^{H \cup H'} \subseteq \mathcal{H}^H \cap \mathcal{H}^{H'}.$$

Because

$$A \cap (H \cup H') = (A \cap H) \cup (A \cap H'),$$

if $A \cap H \in \mathcal{H}$ and $A \cap H' \in \mathcal{H}$, then their union is in $\mathcal{H}$, so $A \in \mathcal{H}^{H \cup H'}$. So combined with the above,

$$\mathcal{H}^{H \cup H'} = \mathcal{H}^H \cap \mathcal{H}^{H'}.$$

Problem 1.1.21. For finite $d \geq 1$, prove both equalities

$$\mathcal{B}_{\mathbb{R}^d} = \mathcal{B}_{\mathbb{R}}^{\otimes d} = \sigma\{(a_1, b_1) \times \cdots \times (a_d, b_d) : a_i < b_i,\ a_i, b_i \in \mathbb{R}\}.$$

Here $\mathcal{B}_{\mathbb{R}}^{\otimes d}$ is the product sigma-algebra generated by Borel rectangles.

Solution.
Define

$$\mathcal{G} = \{(a_1, b_1) \times \cdots \times (a_d, b_d) : a_i < b_i \in \mathbb{R}, i = 1, \ldots, d\}.$$

We first show $\mathcal{B}_{\mathbb{R}^d} = \sigma(\mathcal{G})$. Recall $\mathcal{B}_{\mathbb{R}^d}$ is the σ-algebra generated by the collection of all open sets in $\mathbb{R}^d$. Then $\sigma(\mathcal{G}) \subseteq \mathcal{B}_{\mathbb{R}^d}$ because each open box in $\mathcal{G}$ is an open set in $\mathbb{R}^d$. Also, from the density of $\mathbb{Q}^d$, every open set in $\mathbb{R}^d$ is a countable union of open boxes with rational endpoints, so $\mathcal{B}_{\mathbb{R}^d} \subseteq \sigma(\mathcal{G})$.
Next we show $\sigma(\mathcal{G}) = \mathcal{B} \times \cdots \times \mathcal{B}$. Each open interval (a_i, b_i) of any open box in $\mathcal{G}$ belongs to $\mathcal{B}$, so

$$\mathcal{G} \subseteq \{B_1 \times \cdots \times B_d : B_i \in \mathcal{B}\} \implies \sigma(\mathcal{G}) \subseteq \mathcal{B} \times \cdot \times \mathcal{B}.$$

Since we've shown $\sigma(\mathcal{G}) = \mathcal{B}_{\mathbb{R}^d}$, it sufficies to show that every $B_1 \times \cdots \times B_d \in \mathcal{B} \times \cdots \times \mathcal{B}$ is in $\mathcal{B}_{\mathbb{R}^d}$. For each $i = 1, \ldots, d$, define

$$\mathcal{G}_i = \{B \subseteq \mathbb{R} : \mathbb{R}^{i-1} \times B \times \mathbb{R}^{d-i} \in \mathcal{B}_{\mathbb{R}^d}\}.$$

$\mathcal{G}_i$ is a σ-algebra since $\mathbb{R} \in \mathcal{G}_i$ because $\mathbb{R}^d \in \mathcal{B}_{\mathbb{R}^d}$. Also, if $B \in \mathcal{G}_i$ then $B^c \in \mathcal{G}_i$ because the $\mathbb{R}^{i-1} \times B^c \times \mathbb{R}^{d-i} \in \mathcal{B}_{\mathbb{R}^d}$. And if $B_k \in \mathcal{G}_i$, then $\cup_j B_j \in \mathcal{G}_i$ because

$$\cup_j (\mathbb{R}^{i-1} \times B_j \times \mathbb{R}^{d-i}) = \mathbb{R}^{i-1} \times (\cup_j B_j) \times \mathbb{R}^{d-i}.$$

Lastly, for any open interval $B = (a, b)$ in $\mathbb{R}$, its coordinate cylinder is open in $\mathbb{R}^d$, so $(a, b) \in \mathcal{G}_i$. Since $\mathcal{G}_i$ is a σ-algebra containing all open intervals, $\mathcal{B} \subseteq \mathcal{G}_i$. Thus for any $B_1, \ldots, B_d \in \mathcal{B}$,

$$B_1 \times \cdots \times B_d = \cap_i^d \mathbb{R}^{i-1} \times B_i \times \mathbb{R}^{d-i} \in \mathcal{B}_{\mathbb{R}^d}.$$

So $\mathcal{B} \times \cdots \times \mathcal{B} \subseteq \mathcal{B}_{\mathbb{R}^d} = \sigma(\mathcal{G})$, as desired.

Problem 1.1.22. Let $\mathcal{F} = \sigma(A_\alpha : \alpha \in \Gamma)$ for an uncountable generating family. Prove that every $B \in \mathcal{F}$ belongs to the sigma-algebra generated by some countable subfamily $\{A_{\alpha_j} : j \geq 1\}$.

Solution.
Define

$$\mathcal{G} = \{B \in \sigma(\{A_\alpha : \alpha \in \Gamma\}) : \\ \exists \text{ countable } \{\mathcal{A}_{\alpha_j}\}_{j=1}^\infty \subseteq \{\mathcal{A}_\alpha : \alpha \in \Gamma\} \text{ st. } B \in \sigma(\{\mathcal{A}_{\alpha_j}\}_{j=1}^\infty)\}.$$

Take any $A_\alpha \in \{A_\alpha : \alpha \in \Gamma\}$. Since the singleton $\{A_\alpha\}$ is a countable subcollection and $A_\alpha \in \sigma(\{A_\alpha\})$, we have $A_\alpha \in \mathcal{G}$. Thus $\{A_\alpha : \alpha \in \Gamma\} \subseteq \mathcal{G}$.
We show that $\mathcal{G}$ is closed under complements. Take any $B \in \mathcal{G}$. Then there exists a countalbe subcollection $\{A_{\alpha_j}\}_{j=1}^\infty$ such that $B \in \sigma(\{A_{\alpha_j}\}_{j=1}^\infty)$. Since this is a σ-algebra, $B^c \in \sigma(\{A_{\alpha_j}\}_{j=1}^\infty)$, so $B^c \in \mathcal{G}$.
We also show that $\mathcal{G}$ is closed under countable unions. If $B_i \in \mathcal{G}$, then by definitions, there exists a countable subcollection $\mathcal{C}_i = \{A_{\alpha_{i,j}}\}_{j=1}^\infty \subseteq \{A_\alpha : \alpha \in \Gamma\}$ such that $B_i \in \sigma(\mathcal{C}_i$. Define

$$\mathcal{C} = \cup_i \mathcal{C}_i.$$

A countable union of countable sets is countable, and $\mathcal{C}_i \subseteq \mathcal{C}$ implies $\sigma(\mathcal{C}_i) \subseteq \sigma(\mathcal{C})$. Thus $B_i \in \sigma(\mathcal{C})$ for all i, and since a σ-algebra is closed under countable unions,

$$\cup_i B_i \in \sigma(\mathcal{C}).$$

Therefore $\cup_i B_i \in \mathcal{G}$.

Thus $\mathcal{G}$ is a σ-algebra containing $\{A_\alpha : \alpha \in \Gamma\}$, so

$$\sigma(\{A_\alpha : \alpha \in \Gamma\}) \subseteq \mathcal{G}.$$

By definition, $\mathcal{G} \subseteq \sigma(\{A_\alpha : \alpha \in \Gamma\})$, so

$$\mathcal{G} = \sigma(\{A_\alpha : \alpha \in \Gamma\}).$$

Problem 1.1.33. Show that the algebra assumption in Carathéodory's uniqueness theorem is necessary: construct distinct probability measures μ, ν on $(\Omega, \mathcal{F})$ and a collection $\mathcal{A}$ such that $\sigma(\mathcal{A}) = \mathcal{F}$ and $\mu(A) = \nu(A)$ for all $A \in \mathcal{A}$. Hint: use $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{F} = 2^\Omega$.

Solution.
Consider

$$\mathcal{A} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}\},$$

which generates the singletons of $\mathcal{F}$ using intersections and set subtraction. Take nonequal the probability measures

$$\mu = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right), \qquad v = \left(\frac{1}{8}, \frac{3}{8}, \frac{1}{8}, \frac{3}{8}\right),$$

which are equal for all $A \in \mathcal{A}$:

$$\begin{aligned}
\mu(\{1, 2\}) &= \frac{1}{2} = \nu(\{1, 2\}) \\
\mu(\{2, 3\}) &= \frac{1}{2} = \nu(\{2, 3\}) \\
\mu(\{3, 4\}) &= \frac{1}{2} = \nu(\{3, 4\}).
\end{aligned}$$

Homework 2

Problem 1.2.14. For real random variables $X, X_1, X_2, \ldots$, prove each identity and show that the displayed generating collection is a pi-system:

$$\begin{aligned}
&\text{(a)} \quad \sigma(X) = \sigma(\{X \le a\} : a \in \mathbb{R}), \\
&\text{(b)} \quad \sigma(X_1, \ldots, X_n) = \sigma(\{X_k \le a_k,\ 1 \le k \le n\} : a_1, \ldots, a_n \in \mathbb{R}), \\
&\text{(c)} \quad \sigma(X_1, X_2, \ldots) = \sigma(\{X_k \le a_k,\ 1 \le k \le m\} : m \ge 1,\ a_1, \ldots, a_m \in \mathbb{R}), \\
&\text{(d)} \quad \sigma(X_1, X_2, \ldots) = \sigma\left(\bigcup_{n \ge 1} \sigma(X_1, \ldots, X_n)\right).
\end{aligned}$$

Solution.

(a) Take any A, B in the generating collection of sets. Then $A = \{\omega : X(\omega) \le \alpha\}, \alpha \in \mathbb{R}$ and $B = \{\omega : X(\omega) \le \beta\}, \beta \in \mathbb{R}$. Without loss of generality, assume $\alpha \le \beta$. Then $A \cap B = A$, which is in the generating collection of sets, so it is closed under intersections and is a π-system.

From the definition of a random variable, $\sigma(X)$ is generated by the collection of preimages of the generating collection of sets, which is $X^{-1}((-\infty, \alpha]) = \{\omega : X(w) \le \alpha\}$. Because the intervals $(-\infty, \alpha]$ generate the Borel σ-algebra $\mathcal{B}$, their preimages genereate $\sigma(X)$.

(b) Similarly, take any two A, B in the generating collection of sets. Then $A = \{\omega : X_k(\omega) \le \alpha_k, 1 \le k \le n\}$ and $B = \{\omega : X_k(\omega) \le \beta_k, 1 \le k \le n\}$ and $A \cap B = \{\omega : X_k(\omega) \le \gamma_k, 1 \le k \le n\}$, where $\gamma_k = \min(\alpha_k, \beta_k)$, which is in the generating collection of sets, so it is closed under intersections and is a π-system. With the same reasoning as (a), $\sigma(X_k, 1 \le k \le n)$ is generated by the collection of preimages of the generating collection of sets, which the collection of sets on the right hand side, since they generate $\mathcal{B}_{\mathbb{R}^n}$.

(c) The same argument from (*b*) hold here, since we can take the minimum fo the bounds on the common indices and retain the remaining bounds from the longer set. Then this collection generates $\sigma(X_k, k \le m$ for every $m \in \mathbb{N}$ and from (d), we can see it generates $\sigma(X_1, X_2, \ldots)$.

(d) Take any A, B in the generating collection of sets. Then $A \in \sigma(X_k, k \le n_A)$ and $B \in \sigma(X_k, k \le n_B)$ for some $n_A, n_B \in \mathbb{N}$. Then $A \cap B \in \sigma(X_k, k \le \max(n_A, n_B))$, which is in the countable union.

Because $\sigma(X_1, X_2, \ldots) \supseteq \sigma(X_k, k \le n)$, it must contain their countable union and the σ-algebra generated by that union. Also,

$$\sigma(X_k, k \le n) \subseteq \cup_n \sigma(X_k, k \le n) \subseteq \sigma(\cup_n \sigma(X_k, k \le n)),$$

and $\sigma(X_1, X_2, \ldots)$ is the smallest σ-algebra that makes all X_k measureable, so it must also be a subset of the right hand side. Thus we get equality.

Problem 1.2.20. On a metric space (S, ρ), $g : S \to [-\infty, \infty]$ is lower semicontinuous if $\liminf_{y\to x} g(y) \geq g(x)$ for each x; it is upper semicontinuous if $-g$ is lower semicontinuous. (a) Show that $\{g \leq b\}$ is closed for every real b when g is lower semicontinuous. (b) Deduce that lower and upper semicontinuous functions are Borel measurable. (c) Deduce that continuous functions are Borel measurable.

Solution.

(a) Suppose g is l.s.c. and take any $b \in \mathbb{R}$. Take a sequence of points x_n where $x_n \in \{x : g(x) \leq b\}$. If $x_n \to x$, then because $g(x_n) \leq b$,

$$b \geq \liminf_{n\to\infty} g(x_n) \geq \liminf_{\rho(y,x)\to 0} g(y) \geq g(x),$$

so $x \in \{x : g(x) \leq b\}$ and this set is closed.

(b) From (a), the preimages of $(-\infty, b]$ are $\{x : g(x) \leq b\}$ and closed, which are Borel sets. From Theorem 1.2.9, because $\{(-\infty, b] : b \in \mathbb{R}\}$ generates $\mathcal{B}$, l.s.c. functions are Borel measurable. Similarly, u.s.c. functions are Borel measurable because $-g$ is l.s.c. and their preimages $\{x : -g(x) \leq b\} = \{x : g(x) \geq -b\}$ are closed.

(c) Continuous functions are l.s.c. because

$$\liminf_{\rho(y,x)\to 0} g(y) = \lim_{\rho(y,x)\to 0} g(y) = g(x),$$

so they are Borel measurable.

Problem 1.2.30. Suppose real-valued random variables satisfy $\mathbf{P}(\limsup_n X_n \leq X_\infty) = 1$. Show that for each $\varepsilon > 0$ there exist $A \in \mathcal{F}$ with $\mathbf{P}(A) < \varepsilon$ and a deterministic N such that $X_n(\omega) < X_\infty(\omega) + \varepsilon$ for all $n \geq N$ and every $\omega \notin A$.

Solution.

Choose $\epsilon > 0$ and define $A_N = \cup_{n=N}^{\infty}\{\omega : X_n(\omega) \geq X_\infty(\omega) + \epsilon\}$. It is measurable because it is a countable union of measurable sets. As

$$\cap_{N=1}^{\infty} A_N \subseteq \{\omega : \limsup_{n\to\infty} X_n(\omega) \geq X_\infty(\omega) + \epsilon\},$$

and the RHS has probability zero, $\mathbf{P}(\cap_{N=1}^{\infty} A_N) = 0$. Because probability measures are continuous from above, $\mathbf{P}(A_N) \downarrow 0$. Let $A = A_N$ for large enough N such that $P(A) < \epsilon$, and for any $\omega \in A^c = A_N^c$, $X_n(\omega) < X_\infty(\omega) + \epsilon$ for all $n \geq N$.

Problem 1.2.31. Show that the class $m\mathcal{F}$ of measurable functions is the smallest class containing the simple functions SF and closed under pointwise limits (using the course's measurable-function convention).

Solution.

Lets call the class of functions of interest A. Exercise 1.2.5 shows that all simple functions are measurable and theorem 1.2.22 shows that measureable functions are closed under point-wise limits, so $A \subseteq m\mathcal{F}$.

Prop 1.2.6 shows that for every R.V. $X(\omega) \in m\mathcal{F}$, there exists a sequence of simple functions $X_n(\omega) \to X(\omega)$ for each $\omega \in \Omega$, so $X(\omega) \in A$ and $m\mathcal{F} \subseteq A$.

Problem 1.2.50. For a distribution function F, define $S_F = \{x : F(x+\varepsilon) - F(x-\varepsilon) > 0 \text{ for every } \varepsilon > 0\}$. (a) Show that every discontinuity of F belongs to S_F, and every isolated point of S_F is a discontinuity. (b) Show that S_{F_X} is the support of the law P_X, where a point is in the support of a measure iff every open neighborhood has positive measure.

Solution.

(a) The only discontinuities for non-decreasing functions are jump discontinuities to a higher value. Thus all points of discontinuity must satisfy $F(x+\epsilon) - F(x-\epsilon) > 0$ for all $\epsilon > 0$ all belong to S_F.

Consider any isolated point x in S_F. $F(x+\delta) - F(x-\delta) > 0$, but any $x' \in (x-\delta, x+\delta) \setminus \{x\}$ is not in S_F, so $F(x'+\epsilon) - F(x'-\epsilon) = 0$ for some ϵ implies that F is constant on $(x'-\epsilon, x'+\epsilon)$. Thus F is constant on $(x-\delta, x)$ and $(x, x+\delta)$. Right continuity gives the value at x as the constant on the right. If the left and right constants were equal, F would be constant on a neighborhood of x, contrary to $x \in S_F$. Thus there is a positive jump at x.

(b) Take any x in the support of the law $\mathcal{P}_X$. Then $\mu > 0$ for the open neighborhood $(x-\epsilon, x+\epsilon)$. As $\mathcal{P}_X((x-\epsilon, x+\epsilon]) = F(x+\epsilon) - F(x-\epsilon)$ and this neighborhood has positive measure, the difference must have positive measure and x is in the support of its distribution function F_X.

Take any open neighborhood O around any x in the support of $F(\cdot)$. Choose ϵ such that $(x-\epsilon, x+\epsilon) \subseteq O$ and $\mathcal{P}_X((x-\epsilon/2, x+\epsilon/2]) = F(x+\epsilon/2) - F(x-\epsilon/2) > 0$, so $\mu(O) > 0$ and x is in the support of the law.

Homework 3

Problem 1.3.7. Suppose a probability measure P on $(\mathbb{R}, \mathcal{B})$ satisfies $P(B) = \int f 1_B \, d\lambda$ for all Borel B, where λ is Lebesgue measure and $f \geq 0$ is Borel. Use the standard machine to prove $\int h \, dP = \int fh \, d\lambda$ for every Borel h such that either $h \geq 0$ or $\int f|h| \, d\lambda < \infty$.

Solution.

The first step of the standard machine is to prove this for $h = I_B$. From the assumption,

$$\mathcal{P}(I_B) = \mathcal{P}(B) = \lambda(fI_B).$$

Now take $h \in \mathrm{SF}_+$, $h = \sum_{j=1}^m a_j I_{B_j}, a_j \geq 0$. Linearity shows that

$$\mathcal{P}(h) = \sum_{j=1}^m a_j \mathcal{P}(I_{B_j}) = \sum_{j=1}^m a_j \lambda(fI_{B_j}) = \lambda\left(f \sum_{j=1}^m a_j I_{B_j}\right) = \lambda(fh).$$

For any nonnegative Borel function h, choose nonnegative simple functions $h_n \uparrow h$ where $fh_n \uparrow fh$ because $f \geq 0$. Then,

$$\mathcal{P}(h) \overset{\text{MON}}{=} \lim_n \mathcal{P}(h_n) = \lim_n \lambda(fh_n) \overset{\text{MON}}{=} \lambda(fh).$$

Lastly, suppose h is Borel and $\lambda(f|h|) < \infty$. Write $h = h_+ - h_-$, where because

$$\begin{aligned} \mathcal{P}(h_\pm) = \lambda(fh_\pm) \leq \lambda(f|h|) < \infty, \\ \mathcal{P}(h) = \mathcal{P}(h_+) - \mathcal{P}(h_-) = \lambda(fh_+) - \lambda(fh_-) = \lambda(fh). \end{aligned}$$

Problem 1.3.21. Let $Y \geq 0$ and $v = \mathbf{E}Y^2 < \infty$. (a) For $0 \leq a < \mathbf{E}Y$, prove $\mathbf{P}(Y > a) \geq (\mathbf{E}Y - a)^2 / \mathbf{E}Y^2$ (hint: Cauchy–Schwarz). (b) Prove $(\mathbf{E}|Y^2 - v|)^2 \leq 4v\{v - (\mathbf{E}Y)^2\}$. (c) Prove the second Bonferroni inequality

$$\mathbf{P}\left(\bigcup_{i=1}^n A_i\right) \geq \sum_{i=1}^n \mathbf{P}(A_i) - \sum_{1 \leq j < i \leq n} \mathbf{P}(A_i \cap A_j),$$

and compare it with (a) applied to $Y = \sum_{i=1}^n 1_{A_i}$.

Solution.

(a) For $0 \leq a < \mathbf{E}Y$, the quantity $\mathbf{E}Y - a$ is positive. From the hint, consider

$$\begin{aligned} \mathbf{E}Y = \mathbf{E}[Y I_{Y \leq a}] + \mathbf{E}(Y I_{Y > a}) \leq a + \sqrt{\mathbf{E}Y^2}\sqrt{\mathbf{E}I_{Y>a}} \\ \implies (\mathbf{E}Y - a)^2 \leq \mathbf{E}(Y^2)\mathbf{P}(Y > a) \implies \mathbf{P}(Y > a) \geq \frac{(\mathbf{E}Y - a)^2}{\mathbf{E}(Y^2)}. \end{aligned}$$

(b)

$$\begin{aligned} \mathbf{E}|Y^2 - v| &= \mathbf{E}|(Y - \sqrt{v})(Y + \sqrt{v})| \\ &\leq \sqrt{\mathbf{E}(Y - \sqrt{v})^2}\sqrt{\mathbf{E}(Y + \sqrt{v})^2} \\ &= \sqrt{\mathbf{E}(Y^2 - 2Y\sqrt{v} + v)}\sqrt{\mathbf{E}(Y^2 + 2Y\sqrt{v} + v)} \\ \implies (\mathbf{E}|Y^2 - v|)^2 &\leq \left(2v - 2\sqrt{v}\mathbf{E}Y\right)\left(2v + 2\sqrt{v}\mathbf{E}Y\right) \\ &= 4v(v - (\mathbf{E}Y)^2). \end{aligned}$$

(c) Consider an outcome ω that belongs to k of the A_is. Then

$$I_{\cup_i A_i}(\omega) = \begin{cases} 0 & k = 0 \\ 1 & k \geq 1 \end{cases}, \quad \sum_i I_{A_i}(\omega) = k, \quad \sum_{j<i} I_{A_i \cap A_j}(\omega) = \binom{k}{2},$$

because the number of pairs of events containing ω is k choose 2. If $k \geq 1$,

$$1 - \left(k - \binom{k}{2}\right) = \frac{2}{2} - \frac{2k}{2} + \frac{k(k-1)}{2} = \frac{(k-1)(k-2)}{2} \geq 0,$$

so

$$I_{\cup_i A_i} \geq \sum_i I_{A_i} - \sum_{j<i} I_{A_i \cap A_j}.$$

Both sides are zero if $k = 0$. Using monotonicity and linearity,

$$\mathbf{P}(\bigcup_i A_i) \geq \sum_i \mathbf{P}(A_i) - \sum_{j<i} \mathbf{P}(A_i \cap A_j).$$

We can compare this to the bound in (a) by taking $a = 0$ and writing the RHS as $x - y$. For $Y = \sum_{i=1}^n I_{A_i}$. $\mathbf{E}Y = \sum_i \mathbf{P}(A_i) = x$.

$$\begin{aligned}\mathbf{E}(Y^2) &= \mathbf{E}(\sum_i \sum_j I_{A_i} I_{A_j}) = \sum_i \mathbf{E}(I_{A_i}) + 2\sum_{i<j} \mathbf{E}(I_{A_i} I_{A_j}) \\ &= \sum_i \mathbf{P}(A_i) + 2\sum_{i<j} \mathbf{P}(A_i \cap A_j) = x + 2y \\ \implies \mathbf{P}(\bigcup_i A_i) &\geq \frac{x^2}{x+2y}.\end{aligned}$$

The differences in these bounds are

$$(x-y) - \frac{x^2}{x+2y} = \frac{y(x-2y)}{x+2y},$$

so when $x > 2y$ and $y > 0$, the second Bonferroni inequality is stronger. When $x < 2y$, (a) is stronger.

Problem 1.3.36. Show that $X_n \to 0$ almost surely iff for every $\varepsilon > 0$ there exists an integer n such that every measurable random integer M with $M(\omega) \geq n$ for all ω satisfies $\mathbf{P}(|X_M| > \varepsilon) < \varepsilon$.

Solution.
First, the forward direction. Fix $\epsilon > 0$ and define

$$A_n = \bigcup_{k \geq n} \{\omega : |X_k(\omega)| > \epsilon\}.$$

Because $X_n \xrightarrow{\text{a.s.}} 0$, for almost every ω, only finitely many indices satisfy $|X_k(\omega)| > \epsilon$. Using

$$A_n \downarrow \bigcap_{n=1}^{\infty} A_n, \qquad \mathbf{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = 0,$$

and continuity from above, $\mathbf{P}(A_n) \to 0$. Choose n with $\mathbf{P}(A_n) < \epsilon$. For every random integer $M \geq n$,

$$\{|X_M| > \epsilon\} \subseteq A_n \implies \mathbf{P}(|X_M| > \epsilon) \leq \mathbf{P}(A_n) < \epsilon.$$

For the reverse direction, fix $\epsilon > 0$ and keep A_n. For any $0 < \delta < \epsilon$, choose $n = n(\delta)$ from the hypothesis. Choose $M(\omega)$ as the first index $k \geq n$ for which $|X_k(\omega)| > \epsilon$. If there is no such k then set $M(\omega) = n$. M is measureable because its constructed from measureable events. Then

$$\{\omega : |X_{M(\omega)}| > \epsilon\} = A_n.$$

Since $M(\omega) \geq n$,

$$\mathbf{P}(A_n) = \mathbf{P}(|X_M| > \epsilon) \leq \mathbf{P}(|X_M| > \delta) < \delta.$$

Thus,

$$\mathbf{P}\left(\bigcap_{m=1}^{\infty} A_m\right) \leq \mathbf{P}(A_n) < \delta.$$

Since we can choose δ arbitrarily small, this intersection has probability zero. For $\epsilon = 1/l$, using countable subadditivity,

$$0 = \sum_{l=1}^{\infty} \mathbf{P}\left(\limsup_{n\to\infty} \{|X_n| > 1/l\}\right) \geq \mathbf{P}\left(\bigcup_{l=1}^{\infty} \limsup_{n\to\infty} \{|X_n| > 1/l\}\right).$$

Thus, for almost every ω, and for every $l \geq 1$, there exists a $N_l(\omega)$ such that $|X_n| \leq 1/l$ for all $n \geq N_l$. Because $1/l \to 0$, $X_n \to 0$ almost surely.

Problem 1.3.37. Let Y_n be real random variables and N_k positive-integer-valued random variables on the same probability space. (a) Show $Y_{N_k}(\omega) = Y_{N_k(\omega)}(\omega)$ is measurable. (b) If $Y_n \to Y_\infty$ a.s. and $N_k \to \infty$ a.s., prove $Y_{N_k} \to Y_\infty$ a.s. (c) Give an example with $Y_n \to 0$ in probability and $N_k \to \infty$ a.s., yet $Y_{N_k} = 1$ a.s. for every k. (d) If $Y_n \to Y_\infty$ a.s. and $\mathbf{P}(N_k > r) \to 1$ for every fixed finite r, prove $Y_{N_k} \to Y_\infty$ in probability.

Solution.

(a) To show Y_{N_k} is a random variable, we just need to show that the below is in $\mathcal{F}$ for every Borel set B:

$$\{\omega : Y_{N_k(\omega)}(\omega) \in B\} = \bigcup_{j=1}^{\infty} \left(\{N_k = j\} \cap \{Y_j \in B\}\right).$$

This is true because $\{N_k = j\} \in \mathcal{F}$, $\{Y_j \in B\} \in \mathcal{F}$ and $\mathcal{F}$ is closed under intersections and countable unions.

(b) Suppose $Y_n \xrightarrow{\text{a.s.}} Y_\infty$ and $N_k \xrightarrow{\text{a.s.}} \infty$. Choose ω where both convergences hold. For any $\epsilon > 0$, there exists an N such that

$$n \geq N \implies |Y_n(\omega) - Y_\infty(\omega)| < \epsilon.$$

Similarly, there exists a K such that

$$k \geq K \implies N_k \geq N \implies |Y_{N_k(\omega)}(\omega) - Y_\infty(\omega)| < \epsilon,$$

and $Y_{N_k} \xrightarrow{\text{a.s.}} Y_\infty$.

(c) Consider $\Omega = [0, 1)$ with the uniform probability measure, and let Y_n be the indicators of a sequence of shrinking intervals:

$$[0, 1/2), [1/2, 1) \to [0, 1/4), [1/4, 1/2), [1/2, 3/4), [3/4, 1) \to \cdots.$$

At level k, there are 2^k intervals of length 2^{-k}. For $0 < \epsilon < 1$, the intervals are shrinking with $P(|Y_n| > \epsilon) = 2^{-k(n)}$, where $k(n) \to \infty$ is the level containing interval n, so $Y_n \xrightarrow{p} 0$ ($P(|Y_n| > \epsilon) = 0$ for $\epsilon \geq 1$). For any ω, the index of its interval is $N_k(\omega) = 2^k - 1 + \lfloor 2^k \omega \rfloor \to \infty$, so $Y_{N_k(\omega)}(\omega) = 1$ and $N_k(\omega)$ grows without bound. Thus, $N_k(\omega) \xrightarrow{\text{a.s.}} \infty$.

(d) Take $\epsilon > 0$ and note

$$\{|Y_{N_k} - Y_\infty| > \epsilon\} \subseteq \{N_k < r\} \cup \bigcup_{j \geq r} \{|Y_j - Y_\infty| > \epsilon\},$$

so

$$\mathbf{P}(\{|Y_{N_k} - Y_\infty| > \epsilon\}) \leq \mathbf{P}(\{N_k \leq r\}) + \mathbf{P}\left(\bigcup_{j \geq r} \{|Y_j - Y_\infty| > \epsilon\}\right).$$

For the first term,

$$\mathbf{P}(N_k \leq r) = 1 - \mathbf{P}(N_k > r) \to 0,$$

for every fixed $r < \infty$. Using $Y_n \xrightarrow{\text{a.s.}} Y_\infty$,

$$\mathbf{P}\left(\bigcap_{r=1}^{\infty} \bigcup_{j \geq r} \{|Y_j - Y_\infty| > \epsilon\}\right) = 0 \implies \mathbf{P}\left(\bigcup_{j \geq r} \{|Y_j - Y_\infty| > \epsilon\}\right) \to 0,$$

using continuity from above as $r \to \infty$. Using the inequality we established earlier, taking $k \to \infty$ then $r \to \infty$, we get

$$\mathbf{P}(\{|Y_{N_k} - Y_\infty| > \epsilon\}) \to 0 \implies Y_{N_k} \xrightarrow{\text{p}} Y_\infty.$$

Homework 4

Problem 1.3.65. Write $\mathbb{E}(X; B) = \mathbb{E}[X 1_B]$. Suppose $\mathbb{E}|X| < \infty$, $A_n \in \mathcal{F}$ are disjoint, and $A = \bigcup_{n \geq 0} A_n$. (a) Show $\sum_{n \geq 0} \mathbb{E}(X; A_n)$ converges absolutely and equals $\mathbb{E}(X; A)$. (b) For $Z \geq 0$ with $0 < \mathbb{E}Z < \infty$, deduce that $\mathbf{Q}(B) = \mathbb{E}[Z 1_B]/\mathbb{E}Z$ is a probability measure. (c) Suppose $X, Y \geq 0$, $\mathbb{E}X = \mathbb{E}Y < \infty$, and $\mathbb{E}[X 1_B] = \mathbb{E}[Y 1_B]$ for every B in a pi-system generating $\mathcal{F}$. Prove $X = Y$ a.s.

Solution.

(a) For disjoint A_n and $A = \bigcup_{n\geq 0} A_n$, set $S_N = \sum_{n=0}^{N} X1_{A_n}$. Then $S_N \to X1_A$ pointwise and $|S_N| \leq |X| \in L^1$. By DCT,

$$\mathbb{E}[X1_A] = \lim_{N\to\infty} \sum_{n=0}^{N} \mathbb{E}[X1_{A_n}].$$

Moreover, for each N,

$$\sum_{n=0}^{N} |\mathbb{E}[X1_{A_n}]| \leq \sum_{n=0}^{N} \mathbb{E}[|X|1_{A_n}] \leq \mathbb{E}[|X|1_A] < \infty.$$

Letting $N \to \infty$ proves absolute convergence.

(b) Since $Z \geq 0$ and $0 < \mathbb{E}Z < \infty$, $\mathbf{Q}(B) = \mathbb{E}[Z1_B]/\mathbb{E}Z$ is nonnegative, $\mathbf{Q}(\emptyset) = 0$, and $\mathbf{Q}(\Omega) = 1$. Part (a) gives countable additivity:

$$\sum_n \mathbf{Q}(A_n) = \frac{\sum_n \mathbb{E}[Z1_{A_n}]}{\mathbb{E}Z} = \frac{\mathbb{E}[Z1_A]}{\mathbb{E}Z} = \mathbf{Q}(A).$$

(c) Set $c = \mathbb{E}X = \mathbb{E}Y < \infty$. If $c = 0$, nonnegativity gives $X = Y = 0$ a.s. If $c > 0$, part (b) defines probability measures $\mathbf{R}(B) = \mathbb{E}[X1_B]/c$ and $\mathbf{S}(B) = \mathbb{E}[Y1_B]/c$. They agree on the generating pi-system, so Proposition 1.1.39 gives equality on all of $\mathcal{F}$. Thus $\mathbb{E}[(X-Y)1_B] = 0$ for every $B \in \mathcal{F}$. In particular, the nonnegative variables $(X-Y)1_{\{X>Y\}}$ and $(Y-X)1_{\{Y>X\}}$ both have expectation zero, hence are zero a.s. Therefore $\mathbb{P}(X = Y) = 1$.

Problem 1.4.15. On $\Omega = \{0,1,2\}^2$, give each outcome probability 1/9. Let $X(\omega) = \omega_1$, $Y(\omega) = \omega_2$, $Z_0 = X$, $Z_1 = (X+Y) \bmod 3$, and $Z_2 = (X+2Y) \bmod 3$. (a) Show every pair of Z_0, Z_1, Z_2 is independent, but Z_0, Z_1 determine Z_2. (b) Construct four $\{-1,1\}$-valued variables such that every three are independent but all four are not. Hint: products of independent variables.

Solution.

(a) Each pair of Z_0, Z_1, Z_2 is a bijective linear transformation of (X, Y) mod 3, so each pair is still uniformly distributed and has pairwise independence. But $Z_2 = 2Z_1 - Z_0 \mod 3$, so knowing Z_0, Z_1 determines Z_2.

(b) Let A, B, C be uniform and independent on $\{-1,1\}^3$ and $D = ABC$. Any triple containing D is also uniformly distributed on $\{-1,1\}^3$ since the values determine the other indepedent variable, but $ABCD = 1$ always, so all four are not.

Problem 1.4.18. For $s > 1$, let $\zeta(s) = \sum_{k\geq 1} k^{-s}$ and let X, Y be independent positive-integer-valued random variables with $\mathbb{P}(X = k) = \mathbb{P}(Y = k) = k^{-s}/\zeta(s)$. (a) For each prime p, put $D_p = \{p \mid X\}$ and show these events are mutually independent. (b) Use $\{X = 1\}$ to derive $1/\zeta(s) = \prod_p (1 - p^{-s})$. (c) Show that the probability that no square greater than 1 divides X is $1/\zeta(2s)$. (d) For $G = \gcd(X, Y)$, show $\mathbb{P}(G = k) = k^{-2s}/\zeta(2s)$.

Solution.

(a) Let $p_1, \ldots, p_r$ be distinct primes. Then

$$\mathbb{P}\left(\bigcap_{i=1}^{r} D_{p_i}\right) = \sum_{j=1}^{\infty} \frac{(jp_1 \cdots p_r)^{-s}}{\zeta(s)} = \prod_{i=1}^{r} p_i^{-s} = \prod_{i=1}^{r} \mathbb{P}(D_{p_i}),$$

so the events are mutually independent.

(b) Enumerate the primes as $p_1, p_2, \ldots$. The events $\bigcap_{j=1}^{m} D_{p_j}^c$ decrease to $\{X = 1\}$. Finite independence and continuity from above give

$$\frac{1}{\zeta(s)} = \mathbb{P}(X = 1) = \lim_{m\to\infty} \prod_{j=1}^{m} (1 - p_j^{-s}) = \prod_{p \text{ prime}} (1 - p^{-s}).$$

(c) If no perfect square other than 1 divides X then no p^2 divides X. By the same argument as (a), $\{p^2 | X\}$ for distinct primes p are mutually independent. Thus $\mathbb{P}(p^2|X) = p^{-2s}$ and $\prod_p (1 - p^{-2s}) = \frac{1}{\zeta(2s)}$. Here too the infinite intersection/product identity follows by continuity from above applied to the finite intersections.

(d) Note that $\gcd(X, Y) = k$ iff $X = ka$ and $Y = kb$ for $a, b \in \mathbb{Z}_{\geq 1}$ with $\gcd(a, b) = 1$. Because X and Y are independent,

$$\mathbb{P}(G = k) = \sum_{a,b\geq 1, \gcd(a,b)=1} \mathbb{P}(X = ka, Y = kb) = \sum \frac{(ka)^{-s}(kb)^{-s}}{\zeta(s)^2}$$

$$= k^{-2s} \sum \frac{a^{-s}b^{-s}}{\zeta(s)^2} = k^{-2s}\mathbb{P}(G = 1).$$

Because G is positive,

$$1 = \sum_{k=1}^{\infty} \mathbb{P}(G=k) = \mathbb{P}(G=1)\sum_{k=1}^{\infty} k^{-2s} = \mathbb{P}(G=1)\zeta(2s),$$

so $\mathbb{P}(G=1) = 1/\zeta(2s) \implies \mathbb{P}(G=k) = k^{-2s}/\zeta(2s)$.

Exercises on L_p spaces

1. **Problem L^p part 1.** On a fixed probability space, call X, Y equivalent when $\mathbb{P}(X=Y)=1$. Show that this is an equivalence relation.

   Solution.
   Suppose X and Y are equivalent and also Y and Z are equivalent. Then

   $$\begin{aligned} &\mathbb{P}(\{\omega : X(\omega)=Y(\omega)\}) = 1, \qquad \mathbb{P}(\{\omega : Y(\omega)=Z(\omega)\}) = 1 \\ \implies &\mathbb{P}(\{\omega : X(\omega)=Z(\omega)\}) \geq \mathbb{P}(\{\omega : X(\omega)=Y(\omega)\} \cap \{\omega : Y(\omega)=Z(\omega)\} \\ &= 1 - \mathbb{P}(\{\omega : X(\omega)=Y(\omega)\}^c \cup \{\omega : Y(\omega)=Z(\omega)\}^c) \\ &\geq 1 - \mathbb{P}(\{\omega : X(\omega)=Y(\omega)\}^c) - \mathbb{P}(\{\omega : Y(\omega)=Z(\omega)\}^c) \\ &= 1, \end{aligned}$$

   where equality is satisfied because $\mathbb{P}$ is at most 1. Also,

   $$\begin{aligned} \mathbb{P}(\{\omega : X(\omega)=X(\omega)\}) = \mathbb{P}(\Omega) = 1, \\ \mathbb{P}(\{\omega : X(\omega)=Y(\omega)\}) = \mathbb{P}(\{\omega : Y(\omega)=X(\omega)\}) = 1, \end{aligned}$$

   so we've shown transitivity, reflexivity, and symmetry, and the definition is an equivalence relation.

2. **Problem L^p part 2.** For $p>0$, let L^p consist of equivalence classes with $\mathbb{E}|X|^p < \infty$. Show that $\|X\|_p = (\mathbb{E}|X|^p)^{1/p}$ is a norm when $p \geq 1$. Check well-definedness on the classes, vector-space closure, nonnegativity, definiteness, homogeneity, and the triangle inequality.

   Solution.
   Suppose X and Y in L_p are equivalent, then so must $|X|^p$ and $|Y|^p$. Then they have the same integral and

   $$\mathbb{E}\{|X|^p\} = \int |X(\omega)|^p \, d\mathbb{P}(\omega) = \int |Y(\omega)|^p \, d\mathbb{P}(\omega) = \mathbb{E}\{|Y|^p\},$$

   so $(\mathbb{E}\{|X|^p\})^{1/p} = (\mathbb{E}\{|Y|^p\})^{1/p}$. For any $c \in \mathbb{R}$,

   $$\mathbb{E}|cX|^p = \int |cX(\omega)|^p \, d\mathbb{P}(\omega) = |c|^p \int |X(\omega)|^p \, d\mathbb{P}(\omega) = |c|^p \mathbb{E}|X|^p,$$

   and taking the pth root gives $\|cX\|_p = |c|\|X\|_p$. $|c|^p$ is finite, so $cX \in L_p$. Using $|X(\omega)|, |X(\omega)|^p \geq 0$ and the fact that a nonnegative random variable has expectation zero if and only if it is zero almost surely,

   $$\|X\|_p = 0 \iff \mathbb{E}|X|^p = \int |X(\omega)|^p \, d\mathbb{P}(\omega) = 0,$$

   which is true iff X and 0 are equivalent. Minkowski's inequality shows $\|X+Y\|_p \leq \|X\|_p + \|Y\|_p$, which also implies additive closure, since $p \geq 1$ and $\|X\|_p, \|Y\|_p$ finite.

3. **Problem L^p part 3.** Let L^∞ consist of equivalence classes of random variables bounded almost surely by some finite constant. Show that $\|X\|_\infty = \inf\{M \geq 0 : \mathbb{P}(|X| \leq M) = 1\}$ is a norm on L^∞.

   Solution.
   If $\|X\|_\infty = 0$, then for every $n \geq 1$, $\mathbb{P}(|X| \leq 1/n) = 1$. Using continuity from above,

   $$\mathbb{P}(X=0) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} \{|X| \leq 1/n\}\right) = 1.$$

   Nonnegativity is immediate, and $X=0$ a.s. implies $\|X\|_\infty = 0$. The scalar case $c=0$ is immediate. For any $0 \neq c \in \mathbb{R}$,

   $$\begin{aligned} \|cX\|_\infty &= \inf\{M : \mathbb{P}(|cX| \leq M) = 1\} = \inf\{M : \mathbb{P}(|c||X| \leq M) = 1\} \\ &= \inf\{M : \mathbb{P}(|X| \leq M/|c|) = 1\} \\ &= |c| \inf\{a : \mathbb{P}(|X| \leq a) = 1\} = |c|\|X\|_\infty. \end{aligned}$$

If $|X| \le a$ and $|Y| \le b$ almost surely, then $|X+Y| \le |X| + |Y| \le a+b$ almost surely by the triangle inequality. Thus $\|X+Y\|_\infty \le a+b$ and taking the infimum over all a, b gives

$$\|X+Y\|_\infty \le \|X\|_\infty + \|Y\|_\infty.$$

This also establishes closure under addition. If two random variables are equivalent, their bounds are equal almost surely.

4. **Problem L^p part 4.** For $X \in L^\infty$, show $X \in L^p$ for every finite $p > 0$ and $\|X\|_p \to \|X\|_\infty$ as $p \to \infty$.

Solution.

Define $M = \|X\|_\infty$. If $M = 0$, all the norms are zero, so assume $M > 0$. Because $|X| \le M$ almost surely, $\|X\|_p \le M$, so $X \in L_p$ for every $p > 0$. For any $\epsilon > 0$, define $A_\epsilon = \{|X| > M - \epsilon\}$. Using the essential supremeum, $\mathbb{P}(A_\epsilon) > 0$ when $M > 0$. For $0 < \epsilon < M$,

$$\|X\|_p \ge (M-\epsilon)\mathbb{P}(A_\epsilon)^{1/p} \implies \liminf_{p\to\infty} \|X\|_p \ge M - \epsilon.$$

Taking $\epsilon \to 0$, we get $\lim_{p\to\infty} \|X\|_p = \|X\|_\infty$.

5. **Problem L^p part 5.** Set $S(X) = \mathbb{P}(X \neq 0)$. If $X \in L^q$ for some $q > 0$, prove $\lim_{p \downarrow 0} \|X\|_p^p = S(X)$.

Solution.

If $X \in L_q$ for some $q > 0$, then $|X|^p \to 1_{X \neq 0}$ as $p \to 0$. For $0 < p \le q$, $|X|^p \le 1 + |X|^q$, so we can apply the dominated convergence theorem to get

$$\lim_{p\to 0} \|X\|_p^p = \lim_{p \to 0} \mathbb{E}|X|^p = \mathbb{E}1_{X\neq 0} = \mathbb{P}(X \neq 0) = S(X).$$

6. **Problem L^p part 6.** Show that simple functions are dense in L^p for every $0 < p \le \infty$. For finite p, construct simple X_n with $\mathbb{E}|X_n - X|^p \to 0$; for $p = \infty$, require $\|X_n - X\|_\infty \to 0$. For $0 < p < 1$, use the stated integral criterion rather than calling L^p a normed space. (The explanatory line of the handout prints an unmatched exponent q; the fixed-p criterion above is the intended density statement.)

Solution.

Define the simple function

$$X_n = 2^{-n} \lfloor 2^n X \rfloor 1_{\{|X| \le n\}}.$$

For $|X| \le n$, $|X_n - X| < 2^{-n}$, so

$$\mathbb{E}|X_n - X|^p \le 2^{-np} + \mathbb{E}[|X|^p 1_{\{|X|>n\}}] \to 0,$$

using DCT on the integrable dominating variable $|X|^p$ for the tail term. The truncated, quantized X_n has finitely many values, so it is simple. For the separate case $p = \infty$, assume $X \in L_\infty$ and choose $n > \|X\|_\infty$. Then $|X| \le n$ almost surely, so $\|X_n - X\|_\infty \le 2^{-n} \to 0$.